HDOC Day-53

Aim

To write a C program to find the **maximum sum among all contiguous subarrays of size `k`** using pointers and a separate function `max_subarray_sum()`.

Algorithm

1. Start the program.
2. Read the size of the array.
3. Declare a variable-length integer array `arr[size]`.
4. Read all the elements of the array.
5. Read the value of `k`.
6. Pass the address of the array to `max_subarray_sum()`.
7. Check whether `k` is valid.
    - If `k <= 0` or `k >= size`, display `-1`.
8. Calculate the sum of the first `k` elements.
9. Store this sum in `maxSum`.
10. Find the sum of the remaining contiguous subarrays of size `k`.
11. Compare each sum with `maxSum`.
12. Update `maxSum` whenever a larger sum is found.
13. Display the maximum subarray sum.
14. Stop.

C Program

```
#include <stdio.h>

void max_subarray_sum(int *arr, int size, int k)
{
    int i, sum = 0, maxSum;

    if (k <= 0 || k >= size)
    {
        printf("-1\n");
        return;
    }

    // Calculate sum of first subarray
    for (i = 0; i < k; i++)
    {
```

```
        sum = sum + *(arr + i);
    }

    maxSum = sum;

    // Calculate sum of remaining subarrays
    for (i = k; i < size; i++)
    {
        sum = sum - *(arr + i - k);
        sum = sum + *(arr + i);

        if (sum > maxSum)
        {
            maxSum = sum;
        }
    }

    printf("Maximum subarray sum: %d\n", maxSum);
}

int main()
{
    int size, i, k;

    printf("Enter the size of array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter the array elements: ");
    for (i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter k: ");
    scanf("%d", &k);

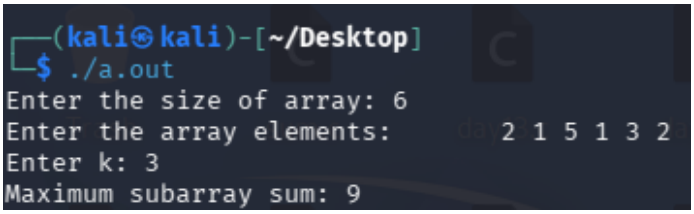
    max_subarray_sum(arr, size, k);

    return 0;
}
```

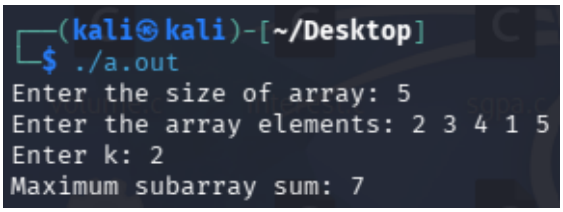
Test Cases

Test Case	Array	k	Output
1	2 1 5 1 3 2	3	Maximum subarray sum: 9
2	2 3 4 1 5	2	Maximum subarray sum: 7
3	1 2 3	4	-1
4	4 2 8 1 6	2	Maximum subarray sum: 10

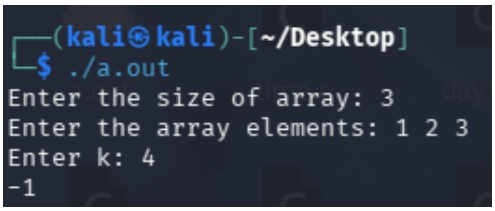
Sample Output 1



Sample Output 2



Sample Output 3 – Invalid k



Result

Thus, the C program to **find the maximum sum among all contiguous subarrays of size `k` using pointers and the `max_subarray_sum()` function** was implemented and executed successfully.